

Reliability / Value-Added / Convenience



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X-Ray Ash Analyzer

LONGi MAGNET CO.,LTD.

The most competitive supplier of industrial magnetic equipment around the world.

ASH ANALYZER

X-ray fluorescence intelligent slurry ash analyzer



Application Scenario:

Used in coal, power, steel, and coking industries for real-time online monitoring of flotation clean coal, providing accurate ash and sulfur data to support production control and quality assurance.

Significance:



Production Control

Real-time ash feedback enables automated process control and feed optimization.



Precise Sulfur Control

Real-time sulfur monitoring optimizes targets while lowering desulfurization costs and environmental risks.



Economic Benefits

Improves yield and quality while reducing manual sampling and testing costs.



Energy Efficiency & Environmental Protection

Optimized flotation lowers water load, emissions, idle operation, and energy consumption.



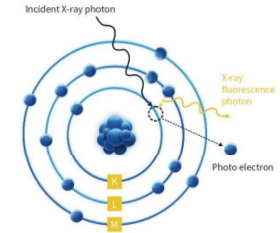
Intelligent Transformation

Supports smart coal preparation policies and advances quality-driven intelligent mining.

Principle:

High-energy X-rays irradiate the sample, causing inner-shell electrons to be ejected and outer-shell electrons to transition and release characteristic X-rays (with element-specific energies). The detector measures the intensity and energy of the fluorescence to analyze elemental composition.

Coal ash content depends on inorganic mineral matter in coal. By measuring the characteristic X-rays of ash-forming elements (such as Si, Al, Fe, Ca) and converting their intensities into elemental concentrations, ash content can be calculated based on composition models.



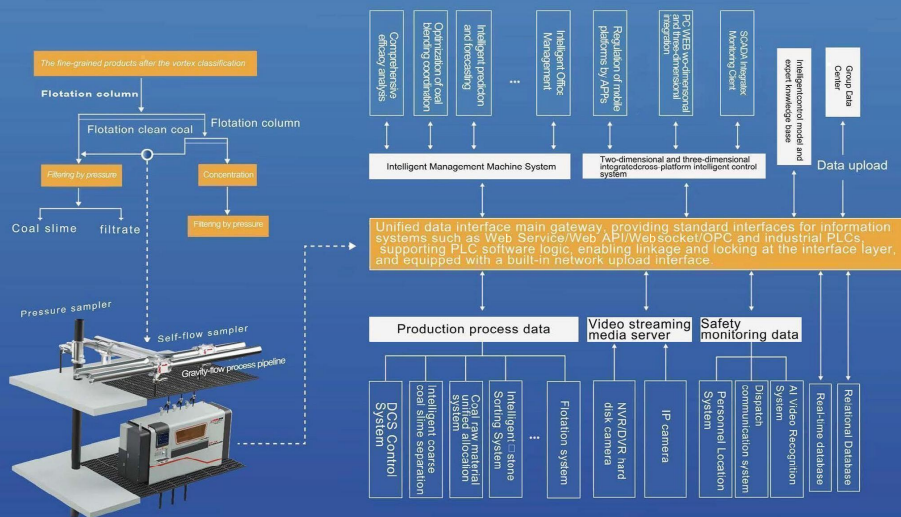
Detection Range

Due to its unique technical capabilities, LXAD can perform in-depth analysis of samples and provides good detection sensitivity for ash-forming elements above Mg, enabling precise quantitative analysis of the ash-forming elements shown in the figure.

Major ash-forming elements Trace ash-forming elements Minor ash-forming elements Detectable range

1 H																	2 He	
3 Li	4 Be							5 B	6 C	7 N	8 O	9 F	10 Ne					
11 Na	12 Mg							13 Al	14 Si	15 P	16 S	17 Cl	18 Ar					
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Uuq	115 Uup	116 Uuh	117 Uus	118 Uuo	
			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu		
			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr		

- Layout diagram



Coal processing expert system

- Product advantages



Strong anti-interference ability and reliable detection:
Not affected by the concentration, speed, bubbles,
stratification, etc. of the coal slurry;



Automatic calibration, precise measurement:
The testing room is equipped with a standard sample station and has an automatic calibration function, eliminating the need for frequent calibration.



One machine, multiple paths, interference isolation:
The sample preparation modules for multiple process coal slurries are independent of each other and operate in parallel, avoiding cross-interference.



Proprietary algorithms, timely maintenance
The data algorithms and analysis software are all independently developed, capable of achieving data transmission over long distances in various production plant modes.

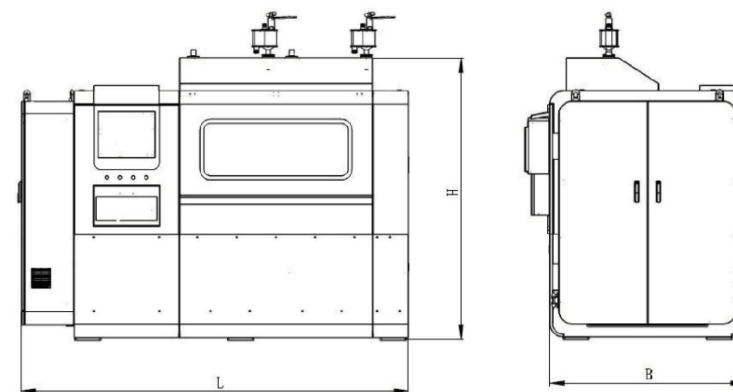


Vacuum detection and interference elimination:
During sample testing, a vacuum treatment is carried out to eliminate the interference of air spectra, effectively improving the analysis efficiency of ash-forming elements and sulfur content.



Sample packaging for easy traceability:
The samples can be automatically packaged at intervals according to the requirements of the selected plant, which is convenient for management and traceability.

- Exterior drawing



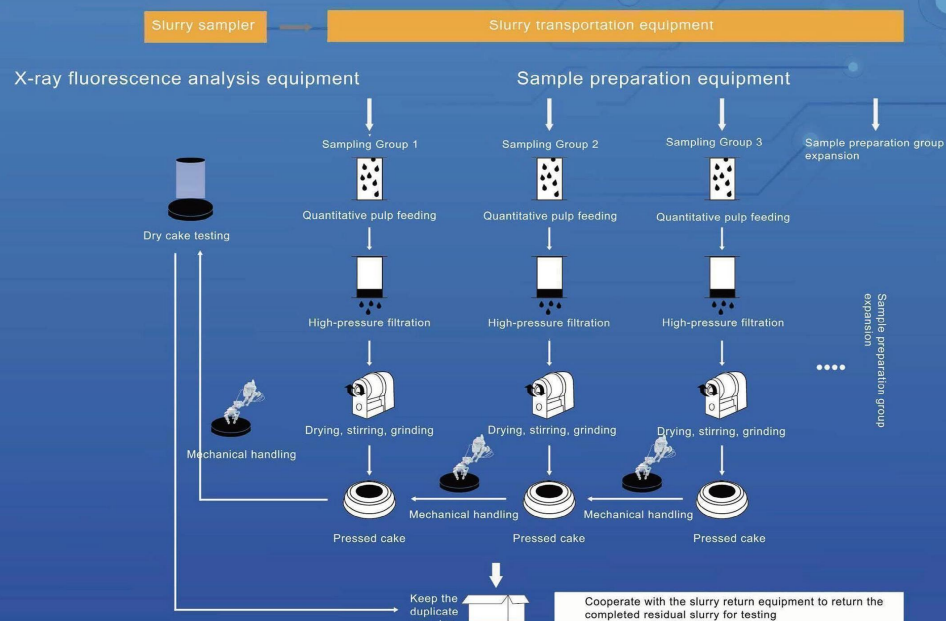
Product Specifications

Item	Dual-Channel	Triple-Channel	unit
Model	LXAD-100-02	LXAD-100-03	—
Power Consumption	8	10	Kw
Power Supply Voltage	AC380		V
Communication Method	Modbus /TCP OPC/HTTP		—
Operating Temperature	0 ~ 50		℃
Ash Measurement Accuracy	Absolute error< ±0.5%		—
Flushing Water Pressure	0.1 ~ 0.5		Mpa
Concentration Range	0 ~ 75%		wt
Detection Time	< 10		Min/time
Detection Frequency	Configure as needed		times/h
Main Unit Dimensions	3000(L)×1100(B)×1900(H)	3500(L)×1100(B)×1900(H)	mm
Main Unit Weight	2200	2600	Kg

Technical route

Sample preparation and analysis process:

Quantitative pulp feeding - High-pressure filtration - Drying, stirring, grinding
- Pressing into dry cakes - Inspection of dry cakes, as shown in the figure below.



System functions



Sensitivity

The analyzer can detect target elements at concentrations as low as 1 % and identify composition changes of 0.01% with 95 % confidence, ensuring reliable analytical accuracy.



Long-term stability

Instrument drift is automatically corrected through built-in calibration. The system continuously compensates for short-term fluctuations, maintaining a stability deviation of less than 0.3%.



Remote data viewing and operation management

The system supports remote monitoring and maintenance services. Users can access ash and sulfur data in real time through web-based and mobile platforms, improving operational efficiency.



Single flow rate

The LXAD analyzer requires a slurry sample flow of 0.5 - 1 L per test, ensuring stable measurement conditions and consistent analytical performance.

Human-machine interface

The product interface is rich and the operation is simple. It can achieve multi-point process control. By setting the sampling interval, continuous analysis for 24 hours can be carried out without interruption. The coal slurry samples taken, the collection volume and the collection frequency are strictly controlled by the program. The detection of ash content values, historical ash content values, ash content curves, reports, etc. can be displayed in real time.



Operating mode



Automatic mode

According to the preset detection frequency Automatic detection



Manual mode

Carry out unplanned single or multiple tests as required for temporary inspections



Maintenance mode

For the equipment's debugging status, the detection data can be manually calibrated separately

APPLICATION CASES

- A Coal Preparation Plant in China



0.4 %

Absolute deviation of ash content

0.1 %

Absolute deviation of sulfur content

6-10 Min

Single channel detection time

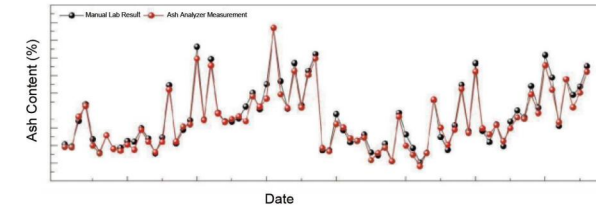
Application Objective:

Introduce an on-line analyzer connected via dual channels to the site intelligent control system. The device monitors composition in real time and sends data to the smart control system, providing timely and accurate composition information to support automated process adjustment.

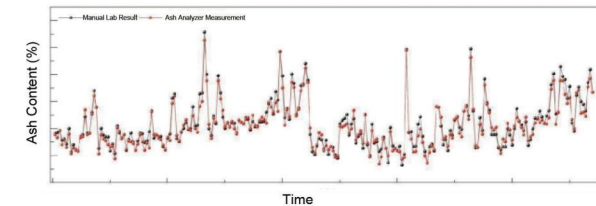
Application Results:

The equipment operates stably, with automatic calibration every two hours. Real-time monitoring cycle: 6–10 minutes; hourly average error $\pm 0.4\%$. Technical indicators meet requirements, and on-site guidance and technical training have been completed.

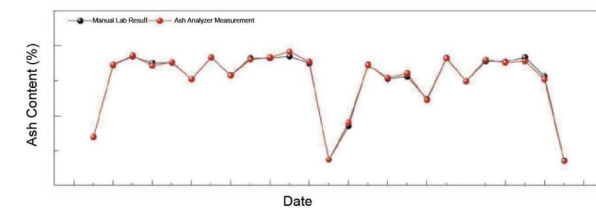
- Daily Ash Content Trend Comparison



- Hourly Ash Content Trend Comparison



- Daily Sulfur Content Trend Comparison



- Hourly Sulfur Content Trend Comparison

